EE2001 Electrical Circuits- I:

Continuation to:

Practices

13 November 2023

Practices

1.26 The numerical values for the currents and voltages in the circuit in Fig. P1.26 are given in Table P1.26. Find the total power developed in the circuit.

Figure P1.26

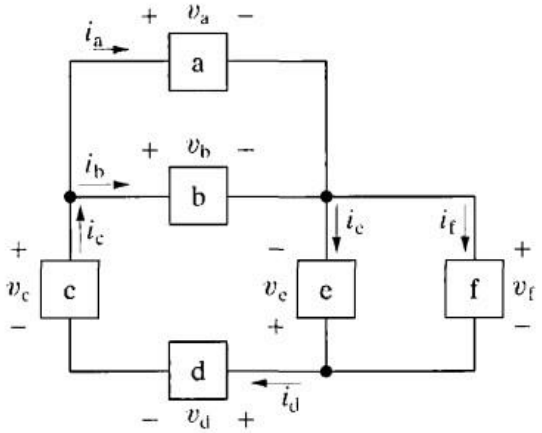


TABLE P1.26

Element	Voltage (kV)	Current (mA)
a	150	0.6
b	150	−1.4
c	100	−0.8
d	250	−0.8
e	300	−2.0
f	−300	1.2

Practices

P 1.26 We use the passive sign convention to determine whether the power equation is $p = vi$ or $p = -vi$ and substitute into the power equation the values for v and i, as shown below:

$$p_a = v_a i_a = (150 \times 10^3)(0.6 \times 10^{-3}) = 90 \text{ W}$$

$$p_b = v_b i_b = (150 \times 10^3)(-1.4 \times 10^{-3}) = -210 \text{ W}$$

$$p_c = -v_c i_c = -(100 \times 10^3)(-0.8 \times 10^{-3}) = 80 \text{ W}$$

$$p_d = v_d i_d = (250 \times 10^3)(-0.8 \times 10^{-3}) = -200 \text{ W}$$

$$p_e = -v_e i_e = -(300 \times 10^3)(-2 \times 10^{-3}) = 600 \text{ W}$$

$$p_f = v_f i_f = (-300 \times 10^3)(1.2 \times 10^{-3}) = -360 \text{ W}$$

Remember that if the power is positive, the circuit element is absorbing power, whereas is the power is negative, the circuit element is developing power. We can add the positive powers together and the negative powers together — if the power balances, these power sums should be equal:

$\sum P_{dev} = 210 + 200 + 360 = 770$ W;

$\sum P_{abs} = 90 + 80 + 600 = 770$ W

Thus, the power balances and the total power developed in the circuit is 770 W.

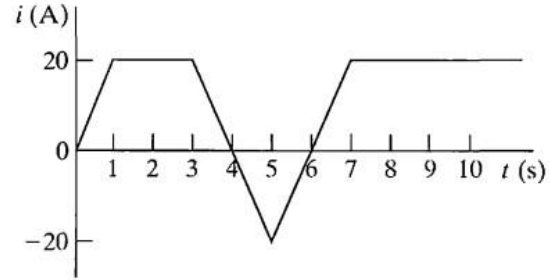
Practices

1.19 The voltage and current at the terminals of the circuit element in Fig. 1.5 are shown in Fig. P1.19.

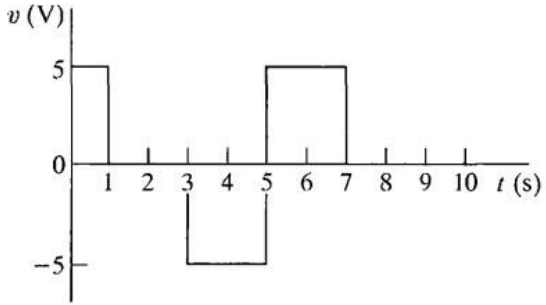
a) Sketch the power versus t plot for $0 \leq t \leq 10$ s.

b) Calculate the energy delivered to the circuit element at $t = 1$, 6, and 10 s.

Figure P1.19



(a)



(b)

Practices

P 1.19 **[a]** $0\text{ s} \le t < 1\text{ s}$:

$v = 5\text{ V};\quad i = 20t\text{ A};\quad p = 100t\text{ W}$

$1\text{ s} < t \le 3\text{ s}$:

$v = 0\text{ V};\quad i = 20\text{ A};\quad p = 0\text{ W}$

$3\text{ s} \le t < 5\text{ s}$:

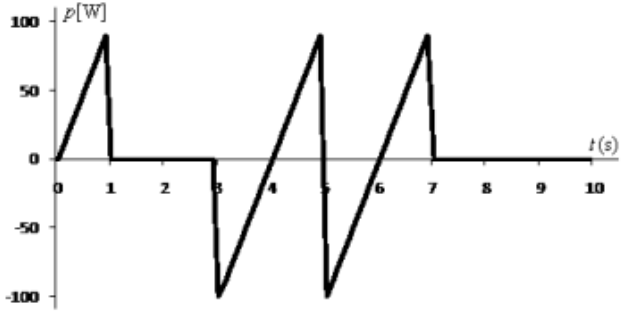
$v = -5\text{ V};\quad i = 80 - 20t\text{ A};\quad p = 100t - 400\text{ W}$

$5\text{ s} < t \le 7\text{ s}$:

$v = 5\text{ V};\quad i = 20t - 120\text{ A};\quad p = 100t - 600\text{ W}$

$t > 7\text{ s}$:

$v = 0\text{ V};\quad i = 20\text{ A};\quad p = 0\text{ W}$



[b] Calculate the area under the curve from zero up to the desired time:

$$w(1) = \tfrac{1}{2}(1)(100) = 50\text{ J}$$

$$w(6) = \tfrac{1}{2}(1)(100) - \tfrac{1}{2}(1)(100) + \tfrac{1}{2}(1)(100) - \tfrac{1}{2}(1)(100) = 0\text{ J}$$

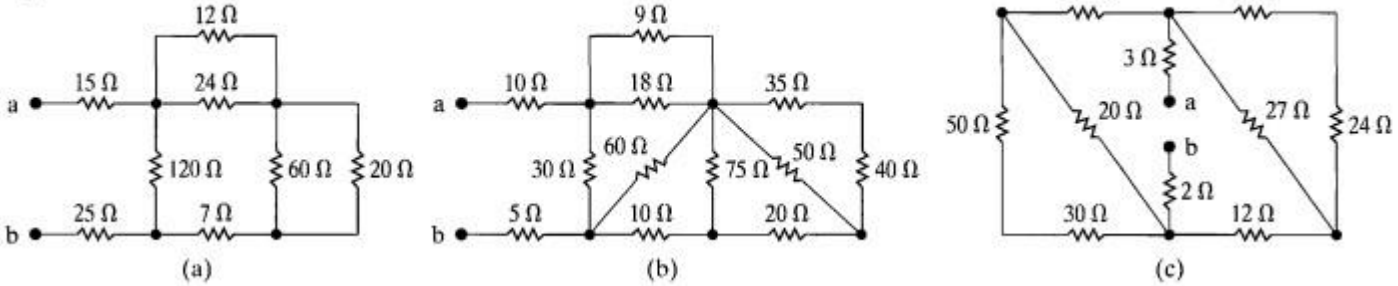
$$w(10) = w(6) + \tfrac{1}{2}(1)(100) = 50\text{ J}$$

Practices

MULTISIM

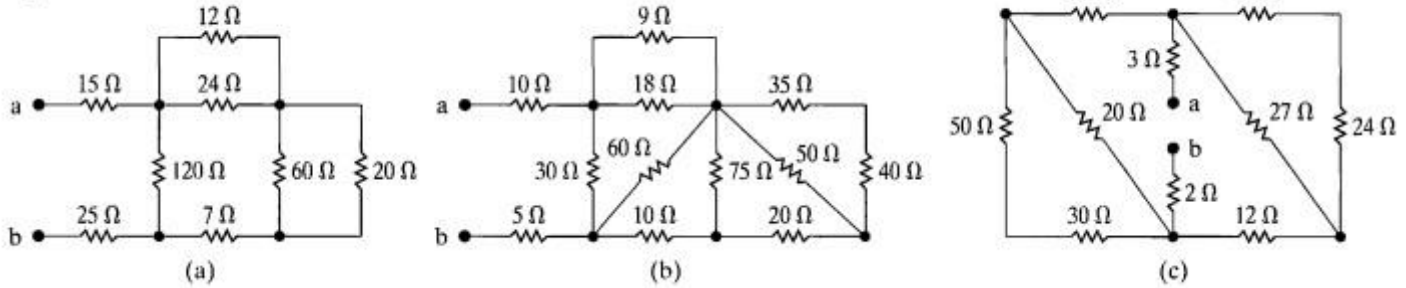
3.6 PSPICE MULTISIM Find the equivalent resistance R_{ab} for each of the circuits in Fig. P3.6.

Figure P3.6



Practices

Figure P3.6



P 3.6 **[a]** $60\|20 = 1200/80 = 15\,\Omega$ $\quad 12\|24 = 288/36 = 8\,\Omega$

$15 + 8 + 7 = 30\,\Omega$ $\quad 30\|120 = 3600/150 = 24\,\Omega$

$R_{ab} = 15 + 24 + 25 = 64\,\Omega$

[b] $35 + 40 = 75\,\Omega$ $\quad 75\|50 = 3750/125 = 30\,\Omega$

$30 + 20 = 50\,\Omega$ $\quad 50\|75 = 3750/125 = 30\,\Omega$

$30 + 10 = 40\,\Omega$ $\quad 40\|60 + 9\|18 = 24 + 6 = 30\,\Omega$

$30\|30 = 15\,\Omega$ $\quad R_{ab} = 10 + 15 + 5 = 30\,\Omega$

[c] $50 + 30 = 80\,\Omega$ $\quad 80\|20 = 16\,\Omega$

$16 + 14 = 30\,\Omega$ $\quad 30 + 24 = 54\,\Omega$

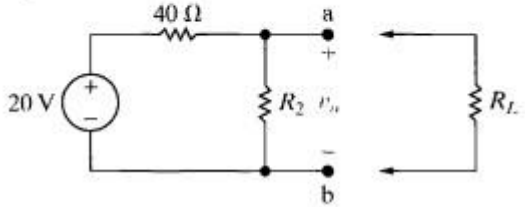
$54\|27 = 18\,\Omega$ $\quad 18 + 12 = 30\,\Omega$

$30\|30 = 15\,\Omega$ $\quad R_{ab} = 3 + 15 + 2 = 20\,\Omega$

Practices

3.14 PSPICE MULTISIM In the voltage-divider circuit shown in Fig. P3.14, the no-load value of v_o is 4 V. When the load resistance R_L is attached across the terminals a and b, v_o drops to 3 V. Find R_L.

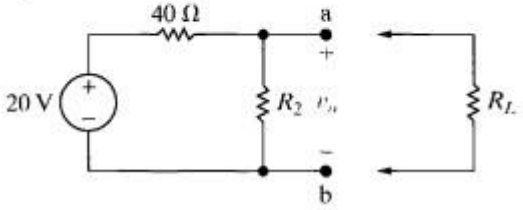
Figure P3.14



Practices

3.14 PSPICE MULTISIM In the voltage-divider circuit shown in Fig. P3.14, the no-load value of v_o is 4 V. When the load resistance R_L is attached across the terminals a and b, v_o drops to 3 V. Find R_L.

Figure P3.14



P 3.14 $$4 = \frac{20R_2}{R_2 + 40} \quad \text{so} \quad R_2 = 10\,\Omega$$

$$3 = \frac{20R_e}{40 + R_e} \quad \text{so} \quad R_e = \frac{120}{17}\,\Omega$$

Thus, $$\frac{120}{17} = \frac{10R_L}{10 + R_L} \quad \text{so} \quad R_L = 24\,\Omega$$

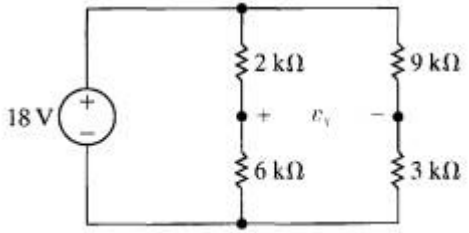
Practices

3.27 PSPICE MULTISIM

a) Find the voltage v_x in the circuit in Fig. P3.27.

b) Replace the 18 V source with a general voltage source equal to V_s. Assume V_s is positive at the upper terminal. Find v_x as a function of V_s.

Figure P3.27



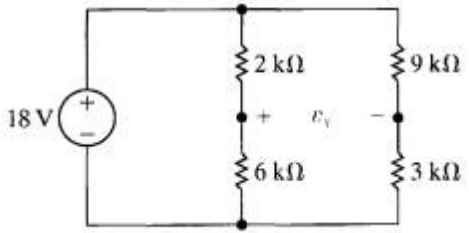
Practices

3.27 PSPICE MULTISIM

a) Find the voltage v_x in the circuit in Fig. P3.27.

b) Replace the 18 V source with a general voltage source equal to V_s. Assume V_s is positive at the upper terminal. Find v_x as a function of V_s.

Figure P3.27



P 3.27 **[a]** $v_{6k} = \dfrac{6}{6+2}(18) = 13.5 \text{ V}$

$$v_{3k} = \frac{3}{3+9}(18) = 4.5 \text{ V}$$

$$v_x = v_{6k} - v_{3k} = 13.5 - 4.5 = 9 \text{ V}$$

[b] $v_{6k} = \dfrac{6}{8}(V_s) = 0.75V_s$

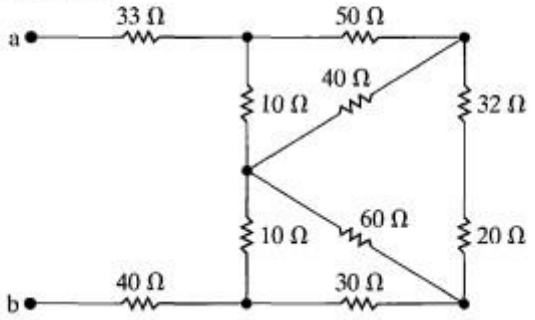
$$v_{3k} = \frac{3}{12}(V_s) = 0.25V_s$$

$$v_x = (0.75V_s) - (0.25V_s) = 0.5V_s$$

Practices

3.58 PSPICE NULTISIM Find the equivalent resistance R_{ab} in the circuit in Fig. P3.58.

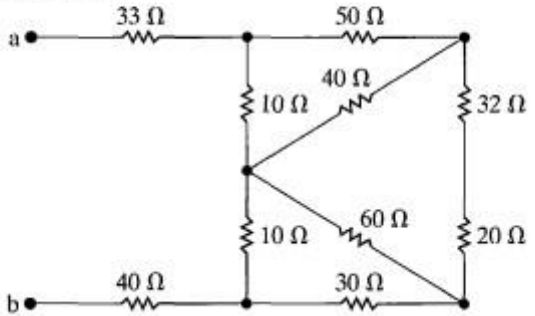
Figure P3.58



Practices

3.58 PSPICE MULTISIM Find the equivalent resistance R_{ab} in the circuit in Fig. P3.58.

Figure P3.58

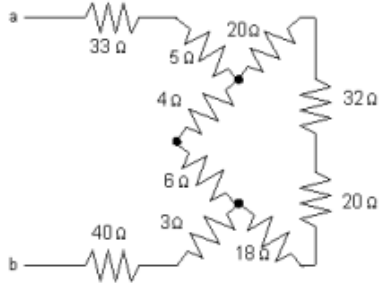


P 3.58 Replace the upper and lower deltas with the equivalent wyes:

$$R_{1U} = \frac{(10)(50)}{100} = 5\,\Omega;\; R_{2U} = \frac{(50)(40)}{100} = 20\,\Omega;\; R_{3U} = \frac{(10)(40)}{100} = 4\,\Omega$$

$$R_{1L} = \frac{(10)(60)}{100} = 6\,\Omega;\; R_{2L} = \frac{(60)(30)}{100} = 18\,\Omega;\; R_{3L} = \frac{(10)(30)}{100} = 3\,\Omega$$

The resulting circuit is shown below:



Now make series and parallel combinations of the resistors:

$$(4+6)\|(20+32+20+18) = 10\|90 = 9\,\Omega$$

$$R_{ab} = 33+5+9+3+40 = 90\,\Omega$$

Practices

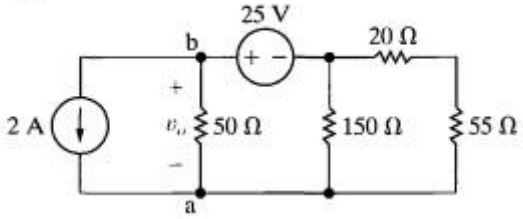
4.26 PSPICE MULTISIM

a) Use the node-voltage method to find v_o and the power delivered by the 2 A current source in the circuit in Fig. P4.26. Use node a as the reference node.

b) Repeat part (a), but use node b as the reference node.

c) Compare the choice of reference node in (a) and (b). Which is better, and why?

Figure P4.26



Practices

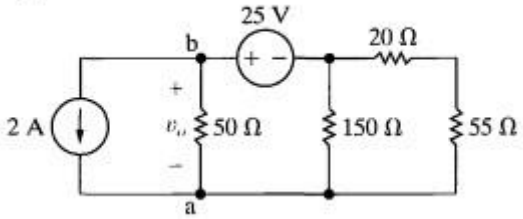
4.26 PSPICE MULTISIM

a) Use the node-voltage method to find v_o and the power delivered by the 2 A current source in the circuit in Fig. P4.26. Use node a as the reference node.

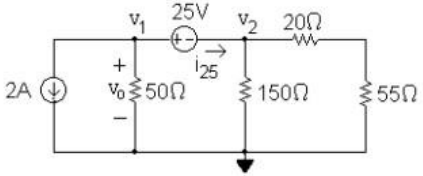
b) Repeat part (a), but use node b as the reference node.

c) Compare the choice of reference node in (a) and (b). Which is better, and why?

Figure P4.26



P 4.26 [a]



This circuit has a supernode includes the nodes v_1, v_2 and the 25 V source. The supernode equation is

$$2 + \frac{v_1}{50} + \frac{v_2}{150} + \frac{v_2}{75} = 0$$

The supernode constraint equation is

$$v_1 - v_2 = 25$$

Place these two equations in standard form:

$$v_1\left(\frac{1}{50}\right) + v_2\left(\frac{1}{150} + \frac{1}{75}\right) = -2$$

$$v_1(1) + v_2(-1) = 25$$

Solving, $v_1 = -37.5$ V and $v_2 = -62.5$ V, so $v_o = v_1 = -37.5$ V.

$$p_{2A} = (2)v_o = (2)(-37.5) = -75 \text{ W}$$

The 2 A source delivers 75 W

Practices

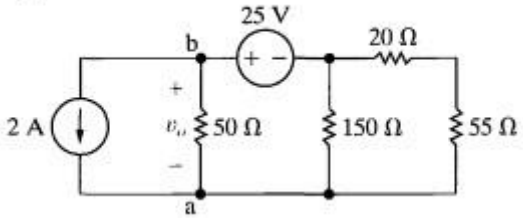
4.26 PSPICE MULTISIM

a) Use the node-voltage method to find v_o and the power delivered by the 2 A current source in the circuit in Fig. P4.26. Use node a as the reference node.

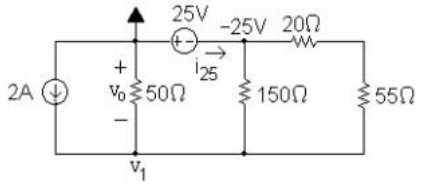
b) Repeat part (a), but use node b as the reference node.

c) Compare the choice of reference node in (a) and (b). Which is better, and why?

Figure P4.26



[b]



This circuit now has only one non-reference essential node where the voltage is not known – note that it is not a supernode. The KCL equation at v_1 is

$$-2+\frac{v_1}{50}+\frac{v_1+25}{150}+\frac{v_1+25}{75}=0$$

Solving, $v_1 = 37.5$ V so $v_0 = -v_1 = -37.5$ V.

$$p_{2A} = (2)v_o = (2)(-37.5) = -75 \text{ W}$$

The 2 A source delivers 75 W.

[c] The choice of a reference node in part (b) resulted in one simple KCL equation, while the choice of a reference node in part (a) resulted in a supernode KCL equation and a second supernode constraint equation. Both methods give the same result but the choice of reference node in part (b) yielded fewer equations to solve, so is the preferred method.

End of the Lesson (13.11.23)

Any questions?

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist